

STED imaging of tau filaments in Alzheimer's disease cortical grey matter



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ABSTRACT

Alzheimer's disease (AD) involves the propagation of filaments of tau protein throughout the cerebral cortex. Imaging tau filaments and oligomers in human brain at high resolution would help contribute insight into the mechanism and progression of tauopathic diseases. STED microscopy is a nano-scale imaging technique and we aimed to test the abilities of this method for resolving tau structures within human brain. Using autopsied 50 μ m AD brain sections, we demonstrate that STED microscopy can resolve immunolabelled tau filaments at 77 nm resolution. Ribbon-like tau filaments imaged by STED appeared smooth along their axis with limited axial undulations. STED also resolved 70–80 nm wide tau puncta. Of the fluorophores tested, STAR635p was optimal for STED imaging in this tissue. This was in part due to brain tissue autofluorescence within the lower wavelength ranges (488–590 nm). Further, the stability and minimal photobleaching of STAR635p allowed STED z-stacks of neurons packed with tau filaments (neurofibrillary tangles) to be collated. There was no loss of x-y image resolution of individual tau filaments through the 20 μ m z-stack. This demonstrates that STED can contribute to nano-scale analysis and characterisation of pathologies within banked human autopsied brain tissue. Resolving tau structures at this level of resolution provides promising avenues for understanding mechanisms of pathology propagation in the different tauopathies as well as illuminating what contributes to disease heterogeneity.

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1. Introduction

Alzheimer's (AD) and other neurodegenerative diseases have characteristic pathologies that are analysed in post-mortem brain collected in clinical and research tissue banks (Augustinack et al., 2002; van der Zee et al., 2008; Holtzman et al., 2012; Wang et al., 2013). Traditional methods of analysis involve immunolabelling and visualization by fluorescence or bright field microscopy. These techniques have a lateral resolution in the order of ~250 nm allowing the sub-cellular localization and overall outline of the pathologies to be visualized. Higher resolution insight into their nanoscale structures and arrangement of their individual components requires transmission electron microscopic (TEM) techniques that are not always routinely possible using post-mortem human tissue processed in brain banks. For this reason, high-resolution optical microscopy methods, like STED (stimulated

emission depletion), for imaging human brain tissue would be of interest in this field. In particular, we focus on the filamentous neural inclusions generated from hyperphosphorylated tau protein that accumulate in hippocampal and cortical projection neurons during the development of AD and other tauopathies (Holtzman et al., 2012; Spillantini and Goedert, 2013).

Tau is an axonal microtubule binding protein that functions to stabilize and regulate cytoskeletal dynamics during development and in synaptically active adult neural cells (Drewes, 2004; Morris et al., 2011; Arendt and Bullmann, 2013). In tauopathic diseases it detaches from axonal microtubules and translocates to the somatodendritic domain where it becomes hyperphosphorylated, self-aggregates and forms neurofibrillary tangles (NFTs) (Kowall and Kosik, 1987; Bancher et al., 1989; Spillantini and Goedert, 2013). Individual tau filaments within NFTs are called PHFs (paired helical filaments) although their molecular structure is more reflective of a straight or twisted ribbon (Barghorn and Mandelkow, 2002; Wang et al., 2013). PHFs visualized by EM are somewhat polymorphic: they have variable lengths of up to several microns, widths varying from 10 to 20 nm, and are sometimes twisted with axial crossover repeats of ~60–80 nm (Barghorn

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and Mandelkow, 2002; Wegmann et al., 2010). Resolving these nanoscale features *in situ* within brain tissue requires visualization by EM. The polymorphic nanoscale structure of tau filaments is important because as well as defining specific forms of tauopathy, it may be essential to the mechanism of tau spread and disease progression. Understanding tau filament structure is therefore relevant to learning more about pathways to disease.

Recent applications of STED microscopy have included super-resolution imaging of components within individual synapses in rodent brain slices (Chereau et al., 2015). New insights from this have included the observation that dendritic spine neck widths change during synaptic potentiation. Optical super-resolution techniques have been evaluated for imaging amyloid-beta fibrils *in vitro* and in cultured cell lines where a large resolution improvement was achieved over conventional optical microscopy, although fibril polymorphisms and fine axial substructure along the fibril axes were not readily resolved (Kaminski Schierle et al., 2011; Pinotsi et al., 2016). There are fewer reports on STED imaging of fixed human tissue. One study showed that STED could resolve ~50 nm structures in immunolabelled human adenocarcinoma sections prepared from paraffin blocks derived from clinical tissue banks (Ilgen et al., 2014). We aimed to evaluate STED compared with conventional confocal microscopy for imaging tau filaments in postmortem human brain sections. Achieving high resolution images of tau filament structure *in situ* within brain tissue could help illuminate processes involved in disease progression and in future possibly highlight differences between types of NFTs and subclasses of tauopathies. We resolve individual tau PHFs within NFTs of human Alzheimer's brain to 77 nm resolution and discuss optimisation of labelling and imaging of these brain sections for possible routine STED microscopy as a technique for neuropathological analysis and research.

2. Materials and methods

2.1. Human tissue immunolabelling

This project was approved by the University of Sydney Human Ethics Committee (protocol number 14432). Sections (50 μ m) of formalin-fixed inferior temporal cortex (ITC) from AD brain were obtained from the NSW Brain Bank (NSWBB). Cases were classified as AD based on neurological dementia diagnosis prior to death and confirmation of AD diagnosis was determined by histopathological examination according to current Australian guidelines for the Consortium to Establish a Registry for Alzheimer's disease (CERAD) criterion and National Institute on Aging (NIA)-Reagan Institute criteria for the neuropathological diagnosis of AD (Harding et al., 2000; Braak et al., 2006; Montine et al., 2012). Sections were immunolabelled with phospho-tau primary antibody 12E8 (Elan Pharmaceuticals) as previously described (Rahman et al., 2014). Different secondary labels were tested for STED including Alexa Fluor 568 goat anti-mouse, Alexa Fluor 647 goat anti-mouse IgG (Life Technologies) and STAR 635P goat anti-mouse IgG (Abberior, 2-0002-007-5). Negative controls comprised sections processed with omission of the primary antibody. In all cases, sections were mounted on slides (Menzel-Glass Ultra Plus) with Prolong Gold anti-fade (Invitrogen P36930).

2.2. STED imaging

STED was performed on a commercial Leica SP5 TCS STED system coupled to an external FLIM module as previously described (Benda et al., 2015). To increase the photostability of the dyes we used the fast 8 kHz resonant scanner combined with anti-fade mounting medium ProlongGold. We used a 100 $\times$ oil immersion

objective (HCX PL APO STED $\times$ 100/1.46; Leica, Wetzlar, Germany) pulsed diode lasers LDH-P-635 and LDH-P-FA-530B driven by PDL 800-B (all PicoQuant, Germany) as excitation sources and a depletion beam is provided by wavelength tunable Ti-Sapphire laser (Mai Tai, Spectra Physics). The excitation power varied from 1 to 20 μ W and the maximum depletion power was 30 mW for 720 nm (Alexa Fluor 568) and 100 mW for 780 nm (Alexa Fluor 647 and STAR 635 P) at the objective back aperture at a frequency of 80 MHz. Synchronization of the excitation and depletion pulses was realized by a software-adjustable delay box (Leica). Minor custom modifications to the excitation path were applied for optimal system performance as described previously (Benda et al., 2015). Emitted fluorescence passed an internal 700 nm short pass filter and an external 590/75 (Alexa Fluor 568) or 680/40 (Alexa Fluor 647 and STAR 635P) bandpass filters (Chroma) and was coupled through an optical fiber to the MicroTime200 main optical unit (PicoQuant, Germany) that contains SPAD detectors (τ -SPAD, PicoQuant, Germany or PDM, MPD, Italy). Single photon detection events were saved in a time-tagged time resolved (TTTR) format with a photon counting module, PicoHarp300 (PicoQuant). In TTTR data, each photon is encoded by its arrival time after the start of the acquisition, its arrival time after a laser excitation pulse and its detector number. Additionally line start markers, provided by the Leica trigger box, were inserted into the TTTR file for image reconstruction. We tested the amount of anti-Stokes excitation at different depletion wavelengths and found out that at 720 nm for Alexa568 and at 780 nm for Alexa647 and STAR-635P the anti-Stokes excitation became negligible (less than 1% of the signal) even for full depletion power. Imaging and setup of scans, including STED alignment, was done using standard LAS AF software. For simultaneous single photon data acquisition in TTTR format and for on-line visualization of intensity time traces and excited state decay histograms, a custom made acquisition program with a user interface was written in LabVIEW (NI). The program uses a dll library provided by PicoQuant to operate the PicoHarp300 photon counting unit. The typical image acquisition setting included pinhole size set to 1 or 0.66 Airy unit, 8 kHz bidirectional scanning, 1024 pixels of size 30 nm, line accumulation 16 and frame accumulation 64 leading to 80 s per image acquisition times. For comparing confocal and STED images, line sequential mode was used.

2.3. Image processing and analysis

Images were reconstructed from the TTTR data using a home built TTTR Data Analysis software (coded in LabVIEW, NI). The reconstructed raw intensity images were exported in TIFF format for further processing and analysis. The software allows automatic correction for sinusoidal scans, phase shift in bidirectional scans and under sampling of line marker timing plus offers photon weighting functionality based on the photon arrival time after excitation, which is typically used for removing the background signal from images. After background subtraction, cross sectional profiles of the smallest visible filaments in the images were analysed using Image J "plot profile" Gaussian "curve fit" tools. The full-width-half-maximum (fwhms) were determined from the Gaussian fits of the plot profiles. 3D animations of STED z-stacks were produced by surface rendering the data using Bitplane Imaris software.

3. Results

3.1. STED imaging of tau filaments at 77 nm resolution in human AD brain sections

Individual tau filaments and bundles were resolved in human brain using STED imaging (Fig. 1a, right). By contrast, this level of

resolution was not seen in confocal images of the same NFTs (Fig. 1a, left). Using labelling with phospho-tau primary antibody and STAR 635P-conjugated secondary IgG antibody, in STED images, the mean width (fwhm) of individual tau filaments was 77 nm (standard deviation 22 nm; $n = 126$) (Fig. 1b–d). By comparison, in confocal images, the mean fwhm of the smallest resolved structures was significantly (three to four fold) larger at 245 nm

(standard deviation 112 nm; $n = 8$; $p < 0.00001$) and individual filaments could not be discerned from bundles (Fig. 1b, c). Substructure along the axis of individual tau filaments was occasionally seen in STED images (Figs. 1 and 2, filaments marked by asterisks). The peak-to-peak spacing of these axial undulations imaged by STED, when visible, were 100–250 nm (Fig. 1e). However, most STED imaged tau filaments in the brain slices appeared relatively

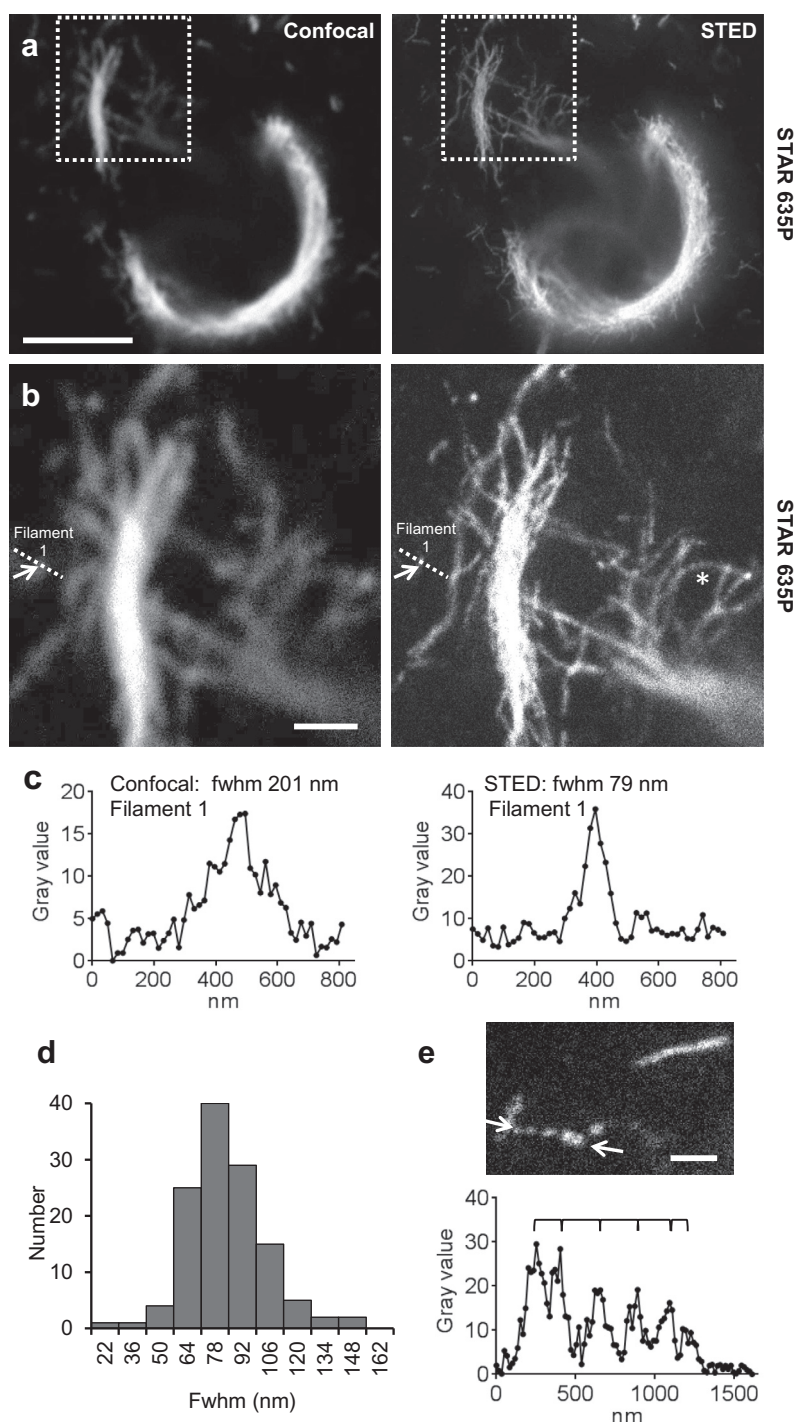


Fig. 1. Human AD brain sections immunolabelled with phospho-tau (12E8) and STAR635P-conjugated secondary antibody. (a) Confocal (left) and STED (right) images of a neurofibrillary tangle. (b) Inset demonstrating substantial improved resolution of individual tau filaments in STED images (right) compared to confocal (left). (c) Example of a cross-sectional profile of an individual tau filament—filament 1 in (b). Measured by STED (right). The fwhm of this filament was 79 nm in the STED image (right) and 201 nm in the confocal image of the same (left). (d) The frequency distribution of fwhms of >100 filaments labelled with the STAR635P fluorophore demonstrated a mean fwhm of 77 ± 2 nm (mean $\pm$ s.e.m.). (e) Axial undulations were visible on some STAR635P labelled filaments by STED. This example of a profile along the filament (arrow to arrow) shows variable axial peak-to-peak spacings of 100–252 nm: Axial peak-peak spacings: 151, 235, 252, 202, 100 nm. The filament at the top left, by contrast, has no obvious axial undulations. Axial undulations, where visible in the images are marked with an asterisk. Scale bars = 5 μm (a) and 1 μm (b, d).

smooth along their axis without resolved substructure (Figs. 1a, b, e and 2). Interestingly, in STED images, dispersed tau filaments were often observed outside of NFTs (Fig. 2a). Some of

these filaments appeared short and fragmented or crossing over each other (Fig. 2b, b', arrow). In addition, tau labelled puncta were also seen inside NFTs as well as more distally (Fig. 2c, c', arrow;

STAR 635P: STED

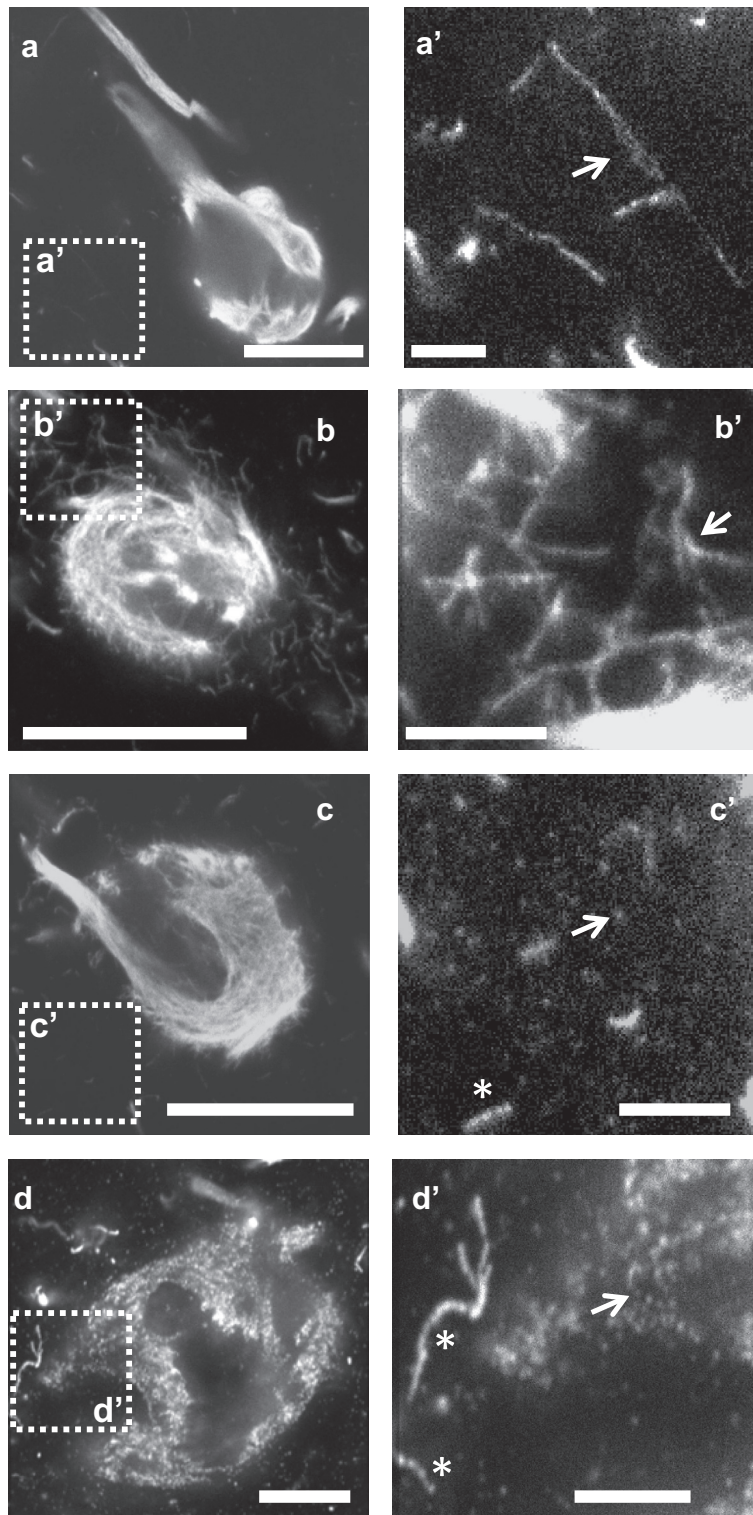


Fig. 2. Structural features of neurofibrillary tangles (left) and higher magnification views of the inset boxes shown (right) imaged with STED using STAR635P. (a, a') Some filaments appeared clearly outside the NFTs (arrow). (b, b') Individual tau filaments were easily discerned within bundles (arrow). (c, c') As well as filaments, smaller punctate tau positive structures were seen outside the NFTs (c', arrow). (d) Tau puncta were also observed intracellularly (d', arrow). Puncta represent positive tau labelling as they were not seen in secondary antibody only control experiments. They were additionally observed using total-tau antibodies (not shown). Axial undulations, where visible in the images are marked with an asterisk. Scale bars = 5 μ m (or 1 μ m insets).

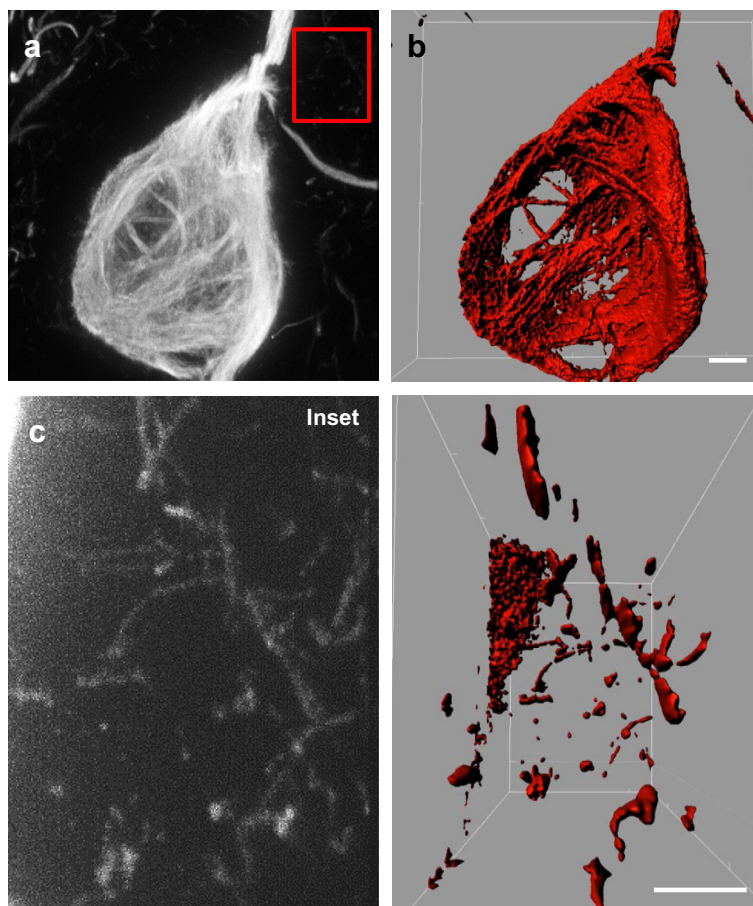
d, d', arrow). Counter-labelling with alpha-interneixin, an intermediate filament and marker of neuronal processes (axons and dendrites; (Benson et al., 1996)), suggested that some of the distal phospho-tau puncta were inside neural processes (Supplementary Fig. 1a, b). Other distal phospho-tau puncta were not inside alpha-interneixin labelled processes (Supplementary Fig. 1a, b) (although we cannot exclude incomplete labelling of axons and dendrites with the intermediate filament marker in autopsied brain tissue). Counter labelling with a total tau antibody showed that almost all of the phospho-tau labelled puncta and larger structures were

also labelled with total tau (Supplementary Fig. 1c). Although not all of the total tau labelled neurites were co-labelled by this phospho-tau antibody (Supplementary Fig. 1d).

3.2. Comparison of fluorophores: STAR 635P enables optimal labelling, imaging and resolution of tau filaments in human brain

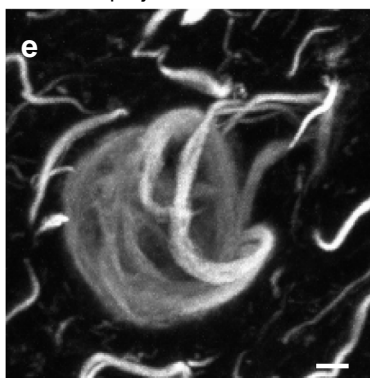
In order to explore an optimised protocol for STED imaging of tau filaments in human post-mortem brain tissue, we additionally compared the use of standard Alexa dyes. We found that Alexa647

STAR 635P: STED



Confocal:

Maximum projection:



Surface-rendered z-stack:

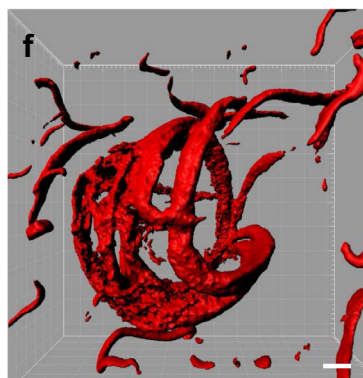


Fig. 3. Surface rendering of STED z-stack of neurofibrillary tangle labelled with STAR635P. (a) Maximum projection of STED stack. (b) 3D surface rendering of z-stack using Imaris. (c) Both tau filaments and puncta are visible in the 3D image (right). (e) Maximum projection confocal image and (f) surface rendering of confocal stack for comparison with the STED data. Scale bars = 2 μ m.

was not optimal for STED as has also previously been observed by others in fluorophore optimisation work (Lau et al., 2011; Benda et al., 2015). The photo-switching behaviour and short lifetime decreasing the fluorescence signal post-depletion, appeared to limit the use of Alexa647 for STED, although this fluorophore does have application in other optical super-resolution techniques (van de Linde et al., 2013). Samples labelled with Alexa647 also displayed significant photobleaching which may be attributed to the *trans-cis* isomerisation typical for carbocyanine dyes, where cis state absorption from singlet and triplet states is red shifted towards 690 nm and thus the cis state of Alexa647 can be very efficiently excited into reactive higher triplet states by the intensive depletion beam (Huang et al., 2006).

Imaging with Alexa568 was better and resulted in relatively low photobleaching compared with Alexa647. Tau filaments labelled with Alexa568-conjugated secondary IgG and imaged by STED had mean widths of 139 nm (standard deviation 39 nm; $n = 43$). This was significantly narrower than the mean width of 255 nm measured in confocal images of Alexa568 labelled tau structures (standard deviation 52 nm; $n = 21$; $p < 0.00001$). Nevertheless, compared to STAR 635P labelled tau, less resolution improvement was achieved. In summary, compared to the Alexa dyes, STAR 635P-conjugated IgG provided the most superior and reproducible high resolution images of individual tau filaments and bundles of filaments within NFTs. Furthermore, fast scanning together with the antifade mounting medium worked very efficiently to suppress photobleaching, to virtually nil, even with the maximal depletion power, for STAR-635P. This allowed 3D imaging using Z-stack acquisitions (Fig. 3). The acquired x-y STED resolution reported above using STAR 635P was hence up to three times that of the confocal with a 0.66 Airy pinhole.

Z-stacks were acquired of sections labelled with phospho-tau and STAR 635P (Fig. 3a, b). The width (fwhm) of tau filaments were measured in individual slices throughout the depth of the 20 μm z-stack. The widths of tau filaments (measured in the x-y plane) of 77 ± 2 nm (mean $\pm$ s.e.m.) did not change significantly from the top to the bottom of stack through this depth of tissue ($p > 0.3$; individual filaments measured from the top ($n = 10$) and from the bottom ($n = 10$) of the stack). Surface rendering of these stacks illustrate the retained resolution in the x-y plane throughout the stack; however, as expected, the z-direction exhibits standard confocal axial resolution resulting in widening of individual filaments in the z-axis (Fig. 3c). The comparative confocal z-stack and surface rendering is shown (Fig. 3e, f).

4. Discussion

Tauopathic diseases involve the hyperphosphorylation and polymerization of the microtubule associated protein tau into filaments that propagate throughout the cerebral cortex (Kruger and Mandelkow, 2015; Lewis and Dickson, 2015). This is a hallmark of several neurodegenerative diseases including Alzheimer's disease, frontotemporal dementia with Parkinsonism linked to chromosome 17 (FTDP-17), progressive supranuclear palsy (PSP) and corticobasal degeneration. Understanding the mechanism of tau filament propagation from cell-to-cell and across brain regions during progression of the different tauopathic diseases would be helped by high resolution imaging of tau inside human brain. Here we provide an optimised protocol for STED imaging of tau filaments in autopsied human brains where we resolved filaments with a mean width of 77 nm as well as small 70–80 nm tau puncta.

In situ structural features including bundling of tau filaments were resolved in STED images. Tau filament fragments and puncta likely representing oligomers are also resolved. The resolving power of STED in this study was three to four fold higher than con-

ventional confocal imaging of the same brain sections. Transmission electron microscopy (TEM) images of tau filaments have revealed that they are often twisted with axial crossover repeats of ~ 60 – 80 nm (Barghorn and Mandelkow, 2002; Wegmann et al., 2010). These cross-over repeats were not visible in STED images – indicating a limitation of this technique. Axial undulations of 100–200 nm were occasionally visible on some filaments in STED images, but we cannot exclude that these may have represented inhomogeneity in the antibody labelling. By contrast in most STED images, individual tau filaments were predominantly smooth along their axis with little undulations to indicate twisting of the filament ribbons as has been observed by TEM and AFM (Barghorn and Mandelkow, 2002; Kumar et al., 2014). It is likely that axial periodicities involving folding of the tau filament ribbon structure require structural resolution within the 5–10 nm range. This is achievable with infiltration of negative stain into the creases within the twisted ribbon and visualization by TEM or imaging using the sub-nm scale resolving power of the AFM probe in the z-direction (Goldsbury et al., 2011; Kumar et al., 2014). Similarly, TEM or AFM is required for resolving substructure along cytoskeletal filaments such as microtubules and F-actin, with these axial nm-scale undulations still being beyond the STED 60–70 nm resolving power (Willig et al., 2006; Yamanaka et al., 2014). Nevertheless, since TEM and AFM are not readily amenable for routine use with fixed autopsied brain sections, STED provides a complementary methodology for high resolution imaging of tau filaments *in situ* within human brain. Different tauopathic diseases and the propagation of tau aggregates throughout the brain have been shown to involve structural polymorphisms in the filaments and oligomers associated with specific diseases and brain regions (Boluda et al., 2015; Lewis and Dickson, 2015). STED level resolution of tau filaments and puncta/oligomers reported here provides the opportunity for further investigations into mechanisms of tau pathology spread and differentiation of the types of filaments and aggregates involved in the different diseases.

Technical aspects of this study have wider relevance to imaging autopsied and immunolabelled thick tissue sections in general. An important aspect of acquiring STED images tens of micrometres deep inside brain or other tissue is the quality of the donut with respect to the refractive index mismatch between the soft tissue and the immersion oil. Mismatch can cause improper focus of both the excitation and depletion beams, which can lead to loss of signal and resolution as imaging proceeds deeper into the tissue. As was demonstrated by Berning and colleagues, for 2D STED using a donut shaped depletion beam, refractive index mismatch deteriorates the quality of the donut, but importantly the central zero is still maintained (Berning, 2011). This means that 2D super-resolution images can be obtained even several tens of micrometres inside the tissue, but apparently with slightly lower lateral resolution (Berning, 2011). To test the effect of imaging depth on achievable resolution in autopsied AD brain slices immunolabelled with tau, we evaluated tau filament profiles at different planes through the z-stack. We did not observe any significant loss of resolution, reaching values of an average of 77 nm for fwhm of tau filament profiles throughout the entire 20 μm z-stack. This may be attributed to the rather low refractive index mismatch between the immersion oil (1.51) and the Prolong Gold mounting medium after several days curing (1.47). Another explanation could be that our resolution is not limited by the STED donut intensity, but by other experimental factors such as scanner uncertainty and/or the physical size of the secondary antibody labelled filaments that we used for testing the resolution. However, STED has been shown to have a level of resolution ranging from 40 to 90 nm even in the presence of primary and secondary antibodies, suggesting that antibody labelling (~ 10 nm) lies in the resolution limit of STED imaging (Kempf et al., 2013; Sidenstein et al., 2016).

STED is an emerging microscopy methodology that has increased applications in achieving super-resolved images in cells, fixed tissue, live tissue and even in live animals. Up until now, studies in human tissue samples have been limited (Ilgen et al., 2014). Recent advances in cultured cell lines include determination of the distribution of membrane proteins within individual mitochondria (Jans et al., 2013). In rodent neurons, STED live-imaging of 40 nm vesicles within synaptic boutons has provided insight into the functional geometry of active synaptic vesicle clusters in cell cultures and more recently *in situ* within rat brain sections (Westphal et al., 2008; Kempf et al., 2013). This has achieved for example identification of different populations of presynaptic vesicles in double labelling experiments (Kempf et al., 2013). STED microscopy of postsynaptic sites demonstrated dynamics in spine neck width in response to synaptic potentiation (Tonnesen et al., 2014). Resolution of the spine neck width was not previously possible with conventional confocal or multiphoton microscopy. Combining time-lapse STED with fluorescence recovery after photobleaching (FRAP) has allowed such morphology changes visualized in conjunction with actual synaptic activity in live brain slices from transgenic mice (Tonnesen and Nagerl, 2013; Tonnesen et al., 2014). In an AD mouse model, STED has demonstrated nanoscale structural degeneration of synapses and corresponding electrophysiological functional outcomes (Siskova et al., 2014). This has correlated nanoscale morphology changes to synaptic hyperexcitability which may be relevant to the earliest manifestation of this neurodegenerative disease (Siskova et al., 2014).

Human brain tissue is available from brain banks for the study of disease mechanisms and for the characterisation and diagnosis of different forms of neurodegenerative disease, although microscopy studies on fixed autopsied human brain samples presents its own set of limitations and difficulties. We have demonstrated here that STED is highly amenable for use on these tissue resources. With the right fluorophores, immunofluorescent labelling and STED allows a mean resolution of 77 nm in this tissue. While this is not enough resolution for fine axial sub-structure along tau filaments to be seen, it does reveal details of the bundling and arrangement of filaments within NFTs and outside NFTs. Therefore, use of STED has the potential to contribute to elucidating significant mechanisms involved in the pathology of neurodegenerative disease. In tau related disease, mechanisms needing further evaluation include the propagation and spread of tau pathology and the cellular heterogeneity and structural differences between the various forms of tauopathy. Resolution of 70–80 nm could also be very useful for addressing further future questions in neuropathology research such as evaluating the possible localization of phospho-tau aggregates to the dendritic spine and synapse. These are areas where the use of STED could contribute.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jsb.2016.07.003>.

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